

# Representational Geometry in Neural Networks

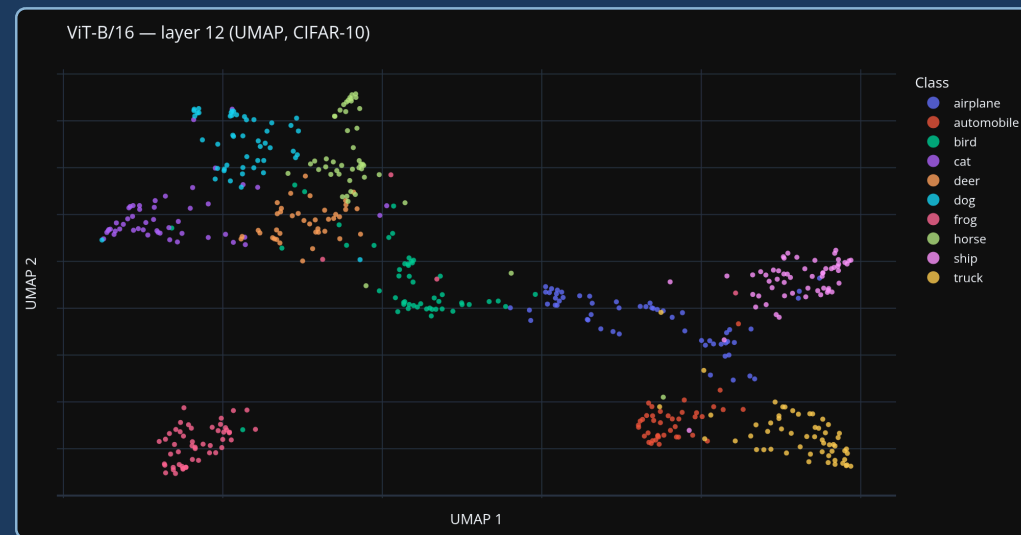
From Vision Transformers to RLHF-Aligned Language Models

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ViT-B/16 · layer 12 · CIFAR-10 · real data

## Part 1 — Cross-modal alignment

When a vision model and a language model train on the **same task**, does a shared representational structure emerge across modalities?

- **Hypothesis A:** cross-modal signal is new information → improves model
- **Hypothesis B:** models converge regardless — extra signal adds little

CKA before/after cross-modal fine-tuning · DINOv2 + Llama  
· MS-COCO

## Part 2 — RLHF geometry

Does RLHF geometrically transform a language model's embedding space — encoding human preference as a **linearly separable structure**?

- Does alignment progressively separate chosen/rejected responses across SFT → DPO → RLHF?
- How does separation evolve **layer by layer**?

UMAP + LinearSVC per layer · Tulu-3-8B SFT/DPO/RLHF · Anthropic/hh-rlhf

Tulu-3-8B SFT → DPO → RLHF · Anthropic/hh-rlhf · 4,000 samples · 33 layers

### Negative finding

Chosen/rejected preference is not linearly encoded in last-token pooled representations, at any alignment stage

- Peak LinearSVC accuracy near chance in all 3 checkpoints — SFT 0.543, DPO 0.544, RLHF 0.550 (layer 14)
- Peak Cohen's d small and flat — 0.183 → 0.193 → 0.194
- No phase transition: alignment does not linearly reorganize preference geometry

### But the geometry does shift

Representational drift (Linear CKA, same inputs, layer by layer):

- CKA(SFT, DPO) drops to **0.976** by layer 32 — most of the shift happens here
- CKA(DPO, RLHF) stays above **0.999** through layer 32 — RLHF/PPO barely moves geometry further
- Shift is concentrated at SFT→DPO — not a chosen/rejected separator, a general representational shift

Full curves (SVC, Cohen's d, CKA drift, anisotropy, effective rank) in the dashboard · Part 2 tab

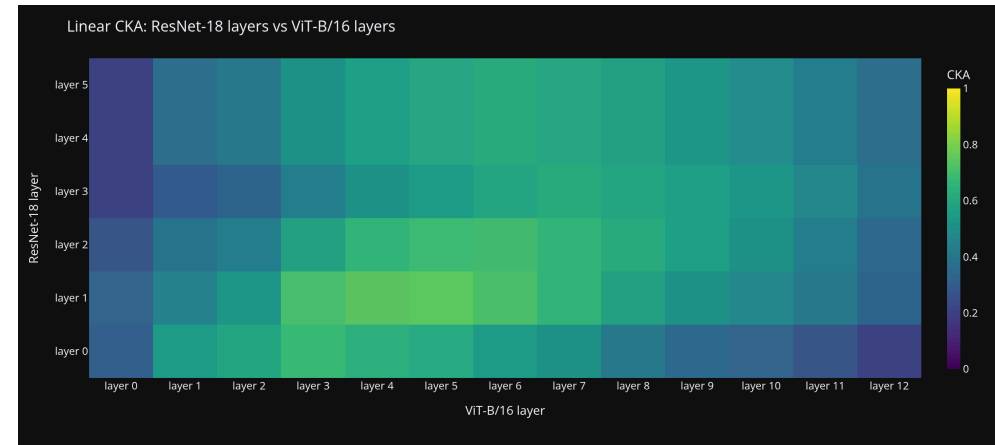
## Setup · DINOv2 + Llama · MS-COCO · multi-class prediction

Both models trained on **same task** (object category prediction) · measure CKA · fine-tune with loss term encouraging similarity to the other model's representation · compare

**CKA high before fine-tuning:** models converge inherently (**Platonic Representation Hypothesis**)

**CKA rises after fine-tuning:** cross-modal signal is genuinely new information

Layer-wise CKA map shows **where** shared meaning lives · UMAP + dashboard to visualize

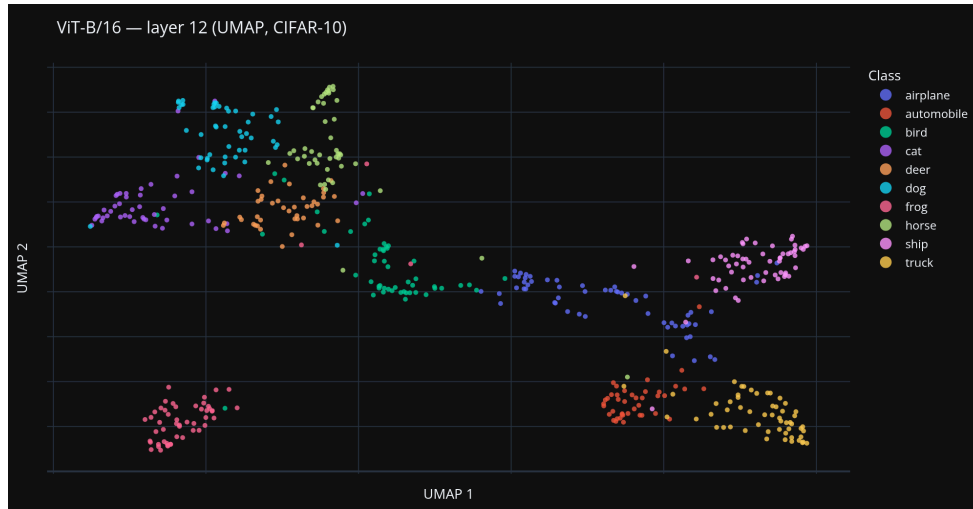


Interactive Plotly Dash · CKA heatmap

## Next steps — toward final

- Cross-modal CKA: DINOv2 vs. Llama on MS-COCO pairs
- **CLIP (ViT-B/32)** as upper bound — explicitly trained cross-modal alignment
- Compare base vs. Instruct CKA scores

# Deliverable & Next Steps



Interactive Plotly Dash · CKA heatmap · UMAP layer slider · LinearSVC score

## Project deliverable

The dashboard **is** the deliverable — interactive visualization of all findings: layer slider, CKA heatmaps, UMAP scatter, SVC score

## Perspective

Extend to RLHF training **snapshots** — plot cross-modal CKA vs. training step to see if alignment shifts **during** RLHF

**He, Trott, Khosla (2025)**

*Shared Latent Representations across Vision and Language*

arXiv:2509.20751 · **Anchor paper**

**Kornblith et al. (2019)**

*Similarity of Neural Network Representations Revisited*

ICML · arXiv:1905.00414 · **CKA method**

**Christiano et al. (2017)**

*Deep RL from Human Preferences*

NeurIPS · arXiv:1706.03741 · **RLHF foundations**

**Lambert et al. (2024)**

*Tulu 3: Pushing Frontiers in Open Language Model Post-Training*

arXiv:2411.15124 · **Models used for Part 2**

**Kucukahmetler et al. (2026)**

*Relative Geometry of Neural Forecasters*

TMLR · arXiv:2602.15676

**Ouyang et al. (2022)**

*Training LMs to Follow Instructions with Human Feedback*

NeurIPS · arXiv:2203.02155 · **InstructGPT / RLHF**

**McInnes et al. (2018)**

*UMAP: Uniform Manifold Approximation and Projection*

arXiv:1802.03426 · **Dim. reduction method**